实验八. 激光雷达避障

实验目的

1）启动激光雷达及了解输出数据流
2）利用激光雷达实现障碍物避障

实验过程

1. 激光雷达类型及参数

型号：VP100 激光雷达

参数	数值
测距原理	三角测距
测量半径	8m
采样频率	4000 次/s
扫描频率	5-10Hz 可调
抗环境光	40Klux
角度分辨率	0.6-1.2° 可调
通信速率	115200bps
最小测量距离	0.12m
尺寸（mm）	94.1*66*40.5
重量	135g

VP100

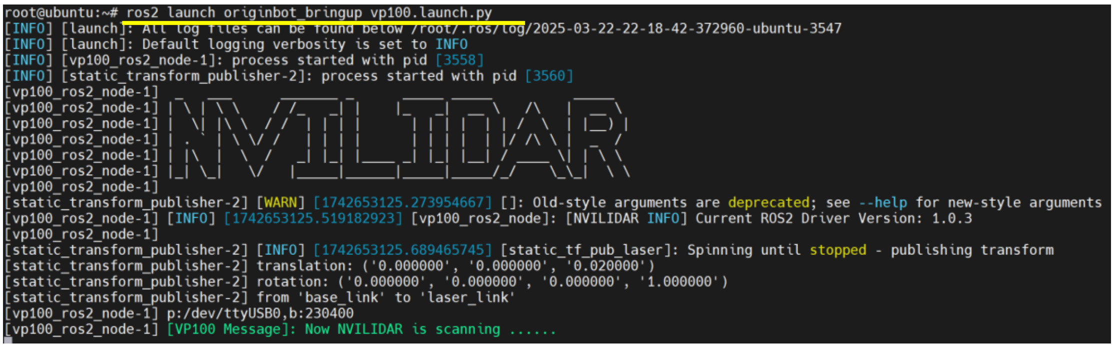
2. 启动并查看激光雷达数据

1）单独启动激光雷达

ros2 launch originbot_bringup vp100.launch.py

或启动底盘及激光雷达

ros2 launch originbot_bringup originbot.launch.py use_lidar:=true

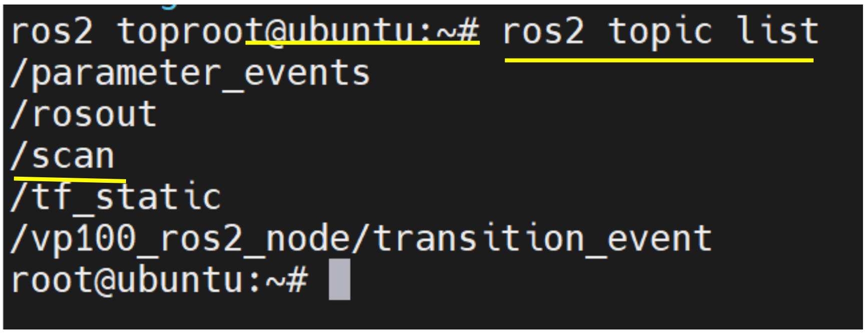


```
root@ubuntu:~# ros2 launch originbot_bringup vp100.launch.py
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-03-22-22-18-42-372960-ubuntu-3547
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [vp100_ros2_node-1]: process started with pid [3558]
[INFO] [static_transform_publisher-2]: process started with pid [3560]
[vp100_ros2_node-1]  _   _ __      __ _____  _       _____  _____             _____  
[vp100_ros2_node-1] | \ | |\ \    / /|_   _|| |     |_   _||  __ \     /\    |  __ \ 
[vp100_ros2_node-1] |  \| | \ \  / /   | |  | |       | |  | |  | |   /  \   | |__) |
[vp100_ros2_node-1] | . ` |  \ \/ /    | |  | |       | |  | |  | |  / /\ \  |  _  / 
[vp100_ros2_node-1] | |\  |   \  /    _| |_ | |____  _| |_ | |__| | / ____ \ | | \ \ 
[vp100_ros2_node-1] |_| \_|    \/    |_____||______||_____||_____/ /_/    \_\|_|  \_\
[vp100_ros2_node-1]
[static_transform_publisher-2] [WARN] [1742653125.273954667] []: Old-style arguments are deprecated; see --help for new-style arguments
[vp100_ros2_node-1] [INFO] [1742653125.519182923] [vp100_ros2_node]: [NVILIDAR INFO] Current ROS2 Driver Version: 1.0.3
[vp100_ros2_node-1]
[static_transform_publisher-2] [INFO] [1742653125.689465745] [static_tf_pub_laser]: Spinning until stopped - publishing transform
[static_transform_publisher-2] translation: ('0.000000', '0.000000', '0.020000')
[static_transform_publisher-2] rotation: ('0.000000', '0.000000', '0.000000', '1.000000')
[static_transform_publisher-2] from 'base_link' to 'laser_link'
[vp100_ros2_node-1] p:/dev/ttyUSB0,b:230400
[vp100_ros2_node-1] [VP100 Message]: Now NVILIDAR is scanning ......
```

2）查看激光雷达数据

新建第二个窗口，查看话题（topic），输入

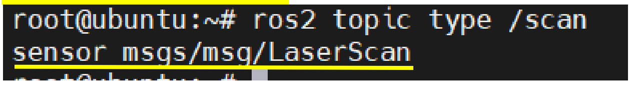
ros2 topic list



```
ros2 toproot@ubuntu:~# ros2 topic list
/parameter_events
/rosout
/scan
/tf_static
/vp100_ros2_node/transition_event
root@ubuntu:~#
```

查看话题类型，输入

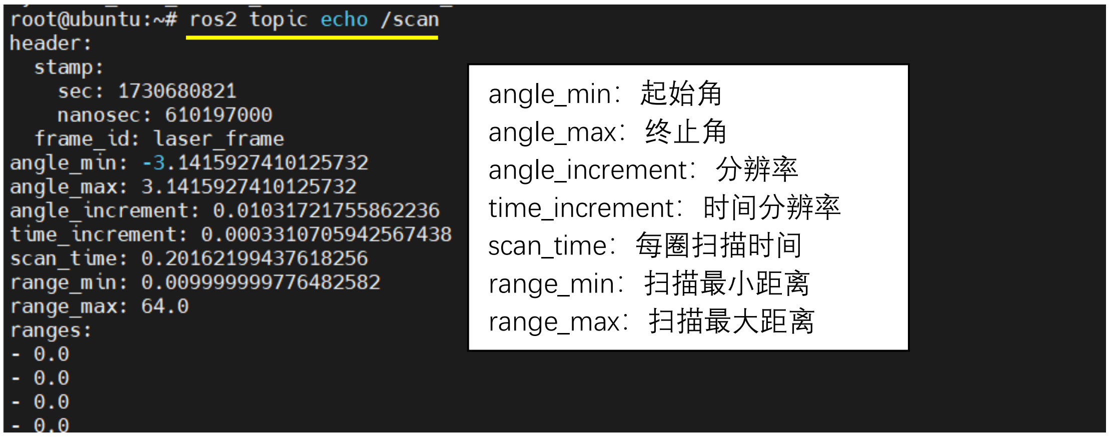
ros2 topic type /scan



```
root@ubuntu:~# ros2 topic type /scan
sensor_msgs/msg/LaserScan
```

查看/scan 数据流，输入后等数据出现后立即 ctr+c 中断

ros2 topic echo /scan



```
root@ubuntu:~# ros2 topic echo /scan
header:
  stamp:
    sec: 1730680821
    nanosec: 610197000
  frame_id: laser_frame
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: 0.010317217558622360
time_increment: 0.0003310705942567438
scan_time: 0.20162199437618256
range_min: 0.009999999776482582
range_max: 64.0
ranges:
- 0.0
- 0.0
- 0.0
- 0.0
```

angle_min：起始角
angle_max：终止角
angle_increment：分辨率
time_increment：时间分辨率
scan_time：每圈扫描时间
range_min：扫描最小距离
range_max：扫描最大距离

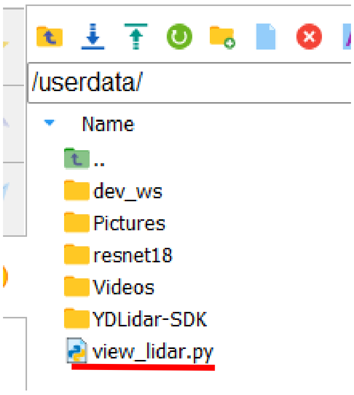
3．查看激光雷达方位角

相关程序 view_lidar.py

```python
#!/usr/bin/env python
# -*- coding: utf-8 -*-
import rclpy
from rclpy.node import Node
from std_msgs.msg import Bool
from sensor_msgs.msg import LaserScan
from rclpy.qos import qos_profile_sensor_data
class ViewLidar(Node):
    def __init__(self,name):
        super().__init__(name)
        self.get_logger().info("LiDAR is OK")
        self.laser_sub= self.create_subscription(LaserScan, "/scan", self.laser_callback, rclpy.qos.qos_profile_sensor_data)
    def laser_callback(self, scan_data):
        for i in range (0, len(scan_data.ranges)):
            if 0.12 < scan_data.ranges[i] <0.3:
                print("ranges["+str(i)+"] = ",scan_data.ranges[i])
```

```
def main(args=None):
    rclpy.init(args=args)
    lidarview = ViewLidar("View_YDLidar")
    rclpy.spin(lidarview)
    lidarview.destroy_node()
    rclpy.shutdown()
if __name__ == '__main__':
    main()
```

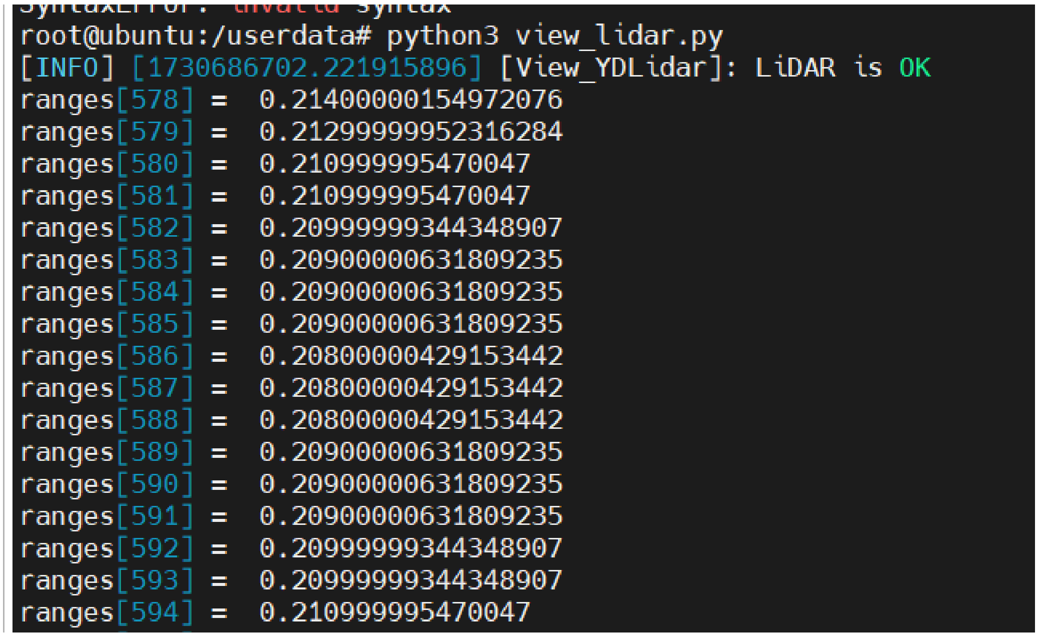
把该程序上传到/userdata 目录下



在第二个窗口运行

```
cd /userdate
python3 view_lidar.py
```

把障碍物靠近激光雷达周围（距离在 0.12-0.3m 间），看显示的 ranges 数值代表方位角



```
SyntaxError: invalid syntax
root@ubuntu:/userdata# python3 view_lidar.py
[INFO] [1730686702.221915896] [View_YDLidar]: LiDAR is OK
ranges[578] =  0.21400000154972076
ranges[579] =  0.21299999952316284
ranges[580] =  0.210999995470047
ranges[581] =  0.210999995470047
ranges[582] =  0.20999999344348907
ranges[583] =  0.20900000631809235
ranges[584] =  0.20900000631809235
ranges[585] =  0.20900000631809235
ranges[586] =  0.20800000429153442
ranges[587] =  0.20800000429153442
ranges[588] =  0.20800000429153442
ranges[589] =  0.20900000631809235
ranges[590] =  0.20900000631809235
ranges[591] =  0.20900000631809235
ranges[592] =  0.20999999344348907
ranges[593] =  0.20999999344348907
ranges[594] =  0.210999995470047
```

4．创建障碍物检测节点并发布是否有障碍物话题（topic）

根据 view_lidar.py 程序，编写 obstacle_detection.py 程序，以达到检测前方障碍物并发布话题功能。

发布是否存在障碍物话题关键语句如下：

```
from std_msgs.msg import Bool
self.pub = self.create_publisher(Bool, '/has_obs', 10)
self.has_obs = Bool()
self.has_obs.data = True    #有障碍物
self.has_obs.data = False    #无障碍物
self.pub.publish(self.has_obs)
```

编写好程序后，

①把障碍物检测 obstacle_detection.py 程序上传到 dev_ws/src/originbot/lanedetection/lanedetection/下

②修改 dev_ws/src/originbot/lanedetection/setup.py

增加一行

```
'obstacle=lanedetection.obstacle_detection:main',
```

③编译

```
cd /userdata/dev_ws
colcon build --packages-select lanedetection --symlink-install
```

5.联调测试

修改 lanedetection.py 文件，把三处注释地方，去掉注释。

```
#self.obs_sub = self.create_subscription(Bool, '/has_obs',self.obs_callback,10)
#self.has_obs = 0
#def obs_callback(self, msg):
        #        self.has_obs = msg.data
```

增加一个判断

```
if self.has_obs:
        小车掉头程序
else:
        原来的车道线程序
self.cmd_vel_pub.publish(self.twist)
```

开 4 个窗口

①`ros2 launch originbot_bringup originbot.launch.py use_camera:=true use_lidar:=true`

②`ros2 run line_follower_perception line_follower_perception`

③`ros2 run lanedetection obstacle`

④`ros2 run lanedetection lanedetection`